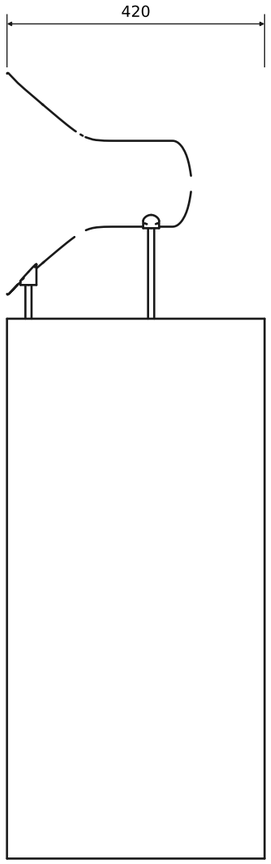
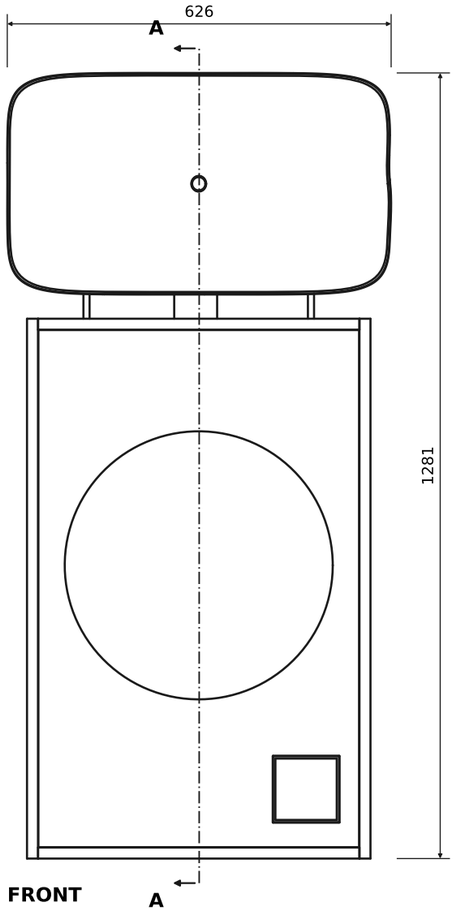
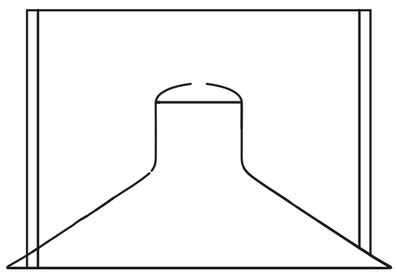
Scale 1:10. First angle: the view from the right is drawn to the left of the front, the view from above under it. A-A: the section on sheet 2.



FROM THE RIGHT



FRONT



FROM ABOVE

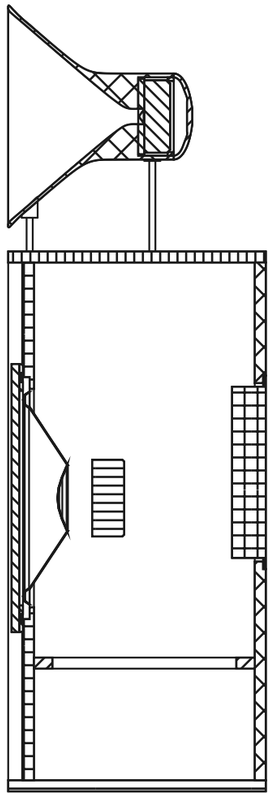
General arrangement

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Section A-A (sheet 1): the product cut at x 280.7 and seen from the right; cut faces hatched, a different angle for each part.



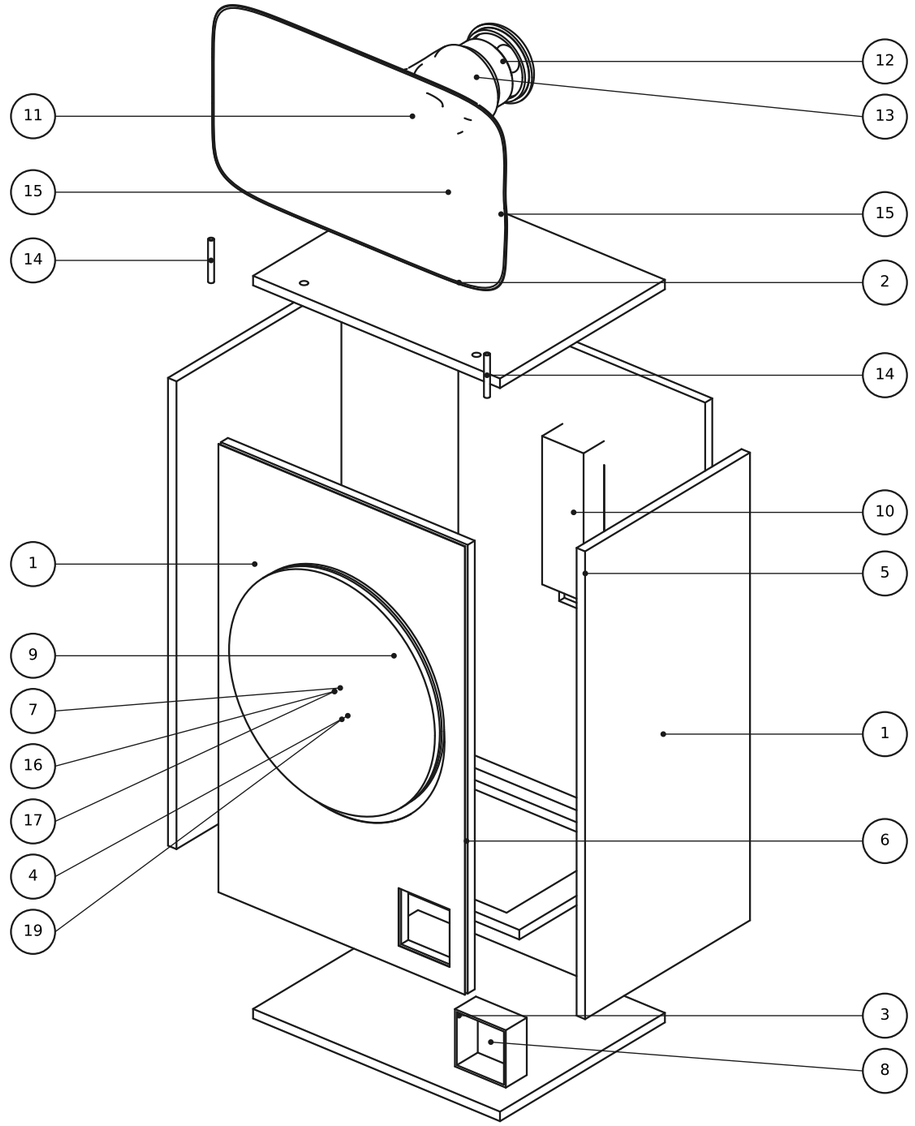
Section A-A

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10, parts drawn apart. Balloons are the parts list's numbers (last sheet).



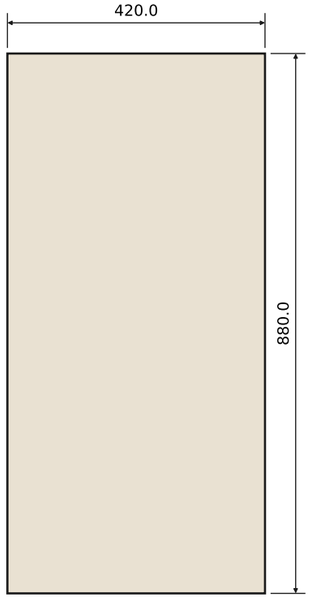
Exploded view

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

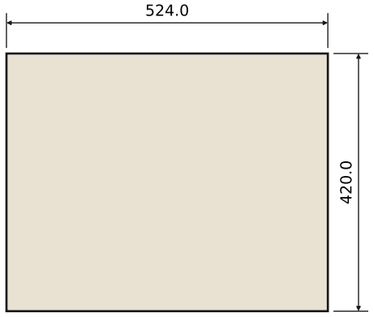
Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:10. Panels cut to their outline only (DXF files in dxf/); sizes as seen from the outer face.



left, right

18 mm baltic-birch, 4 for the set



bottom

18 mm baltic-birch, 2 for the set

Sheet parts: plain panels

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

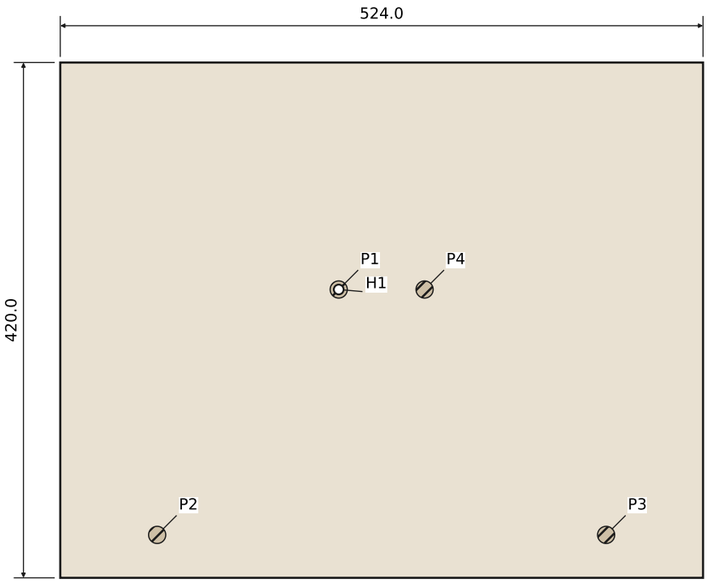
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/top.dxf. Marks listed at the right.

mark	size	x	y	depth
P1	Ø14.0	227.0	235.0	15 from face
P2	Ø14.0	79.0	35.0	15 from face
P3	Ø14.0	445.0	35.0	15 from face
P4	Ø14.0	297.0	235.0	15 from face
H1	Ø8.0	227.0	235.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



top: 18 mm baltic-birch, 2 for the set, seen from its outer face

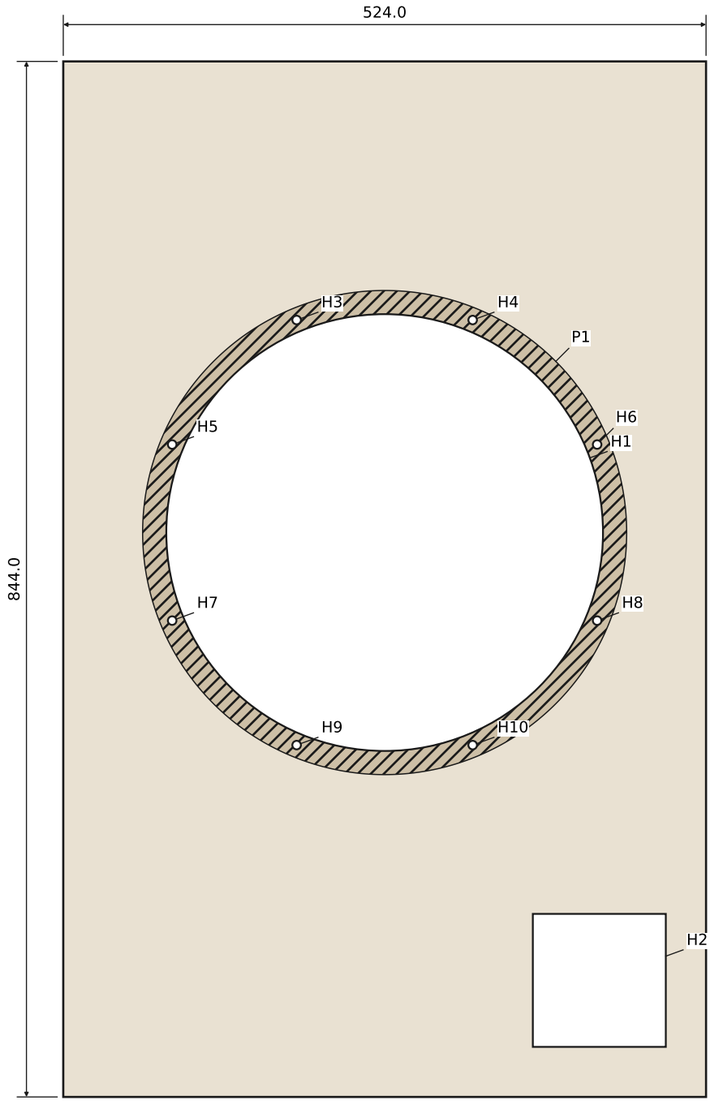
Sheet part: top

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/front-panel.dxf. Marks listed at the right.



front-panel: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	Ø394.6	262.0	460.0	11 from face
H1	Ø356.0	262.0	460.0	through
H2	108.4 x 108.4	437.0	95.0	through
H3	Ø7.0	190.2	633.2	through
H4	Ø7.0	333.8	633.2	through
H5	Ø7.0	88.8	531.8	through
H6	Ø7.0	435.2	531.8	through
H7	Ø7.0	88.8	388.2	through
H8	Ø7.0	435.2	388.2	through
H9	Ø7.0	190.2	286.8	through
H10	Ø7.0	333.8	286.8	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

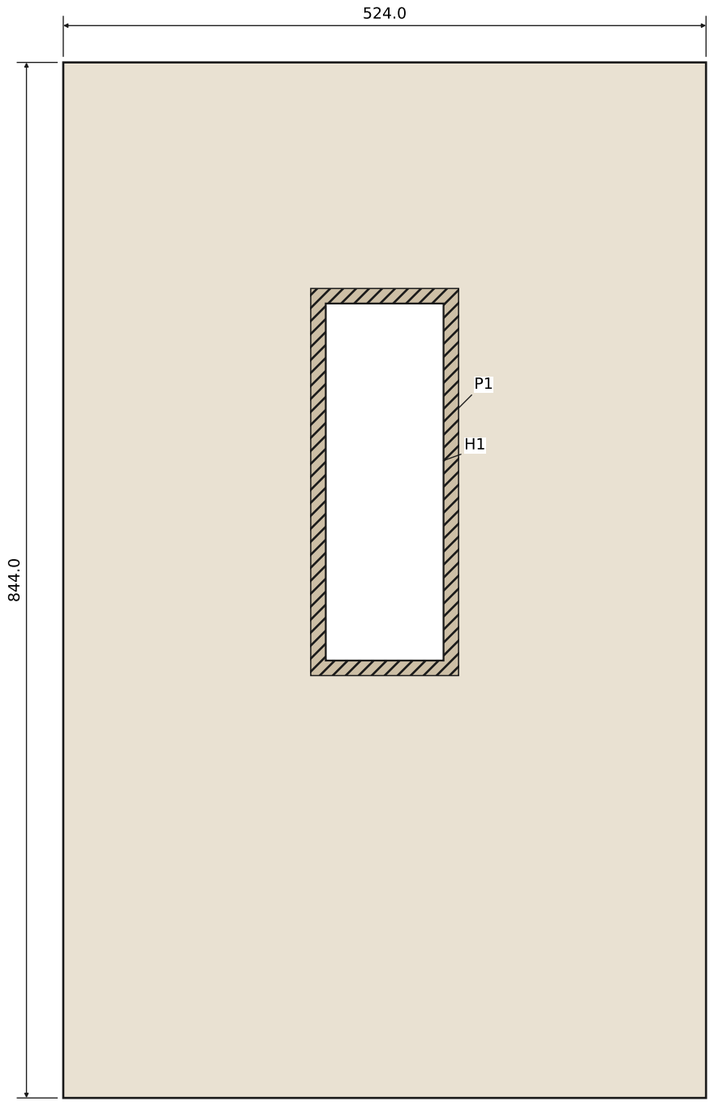
Sheet part: front-panel

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/back.dxf. Marks listed at the right.



back: 18 mm baltic-birch, 2 for the set, seen from its outer face

mark	size	x	y	depth
P1	120.5 x 315.5	262.0	502.0	4.5 from face
H1	96.0 x 291.0	262.0	502.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

Sheet part: back

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

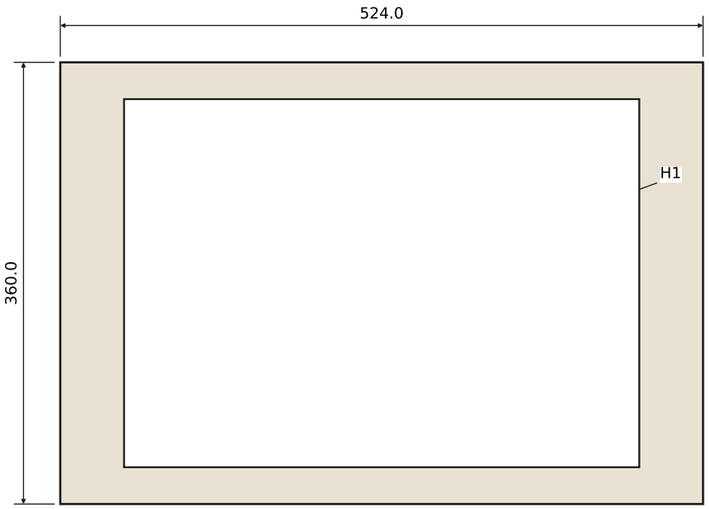
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/brace.dxf. Marks listed at the right.

mark	size	x	y	depth
H1	420.0 x 300.0	262.0	180.0	through

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.



brace: 18 mm baltic-birch, 2 for the set, seen from its outer face

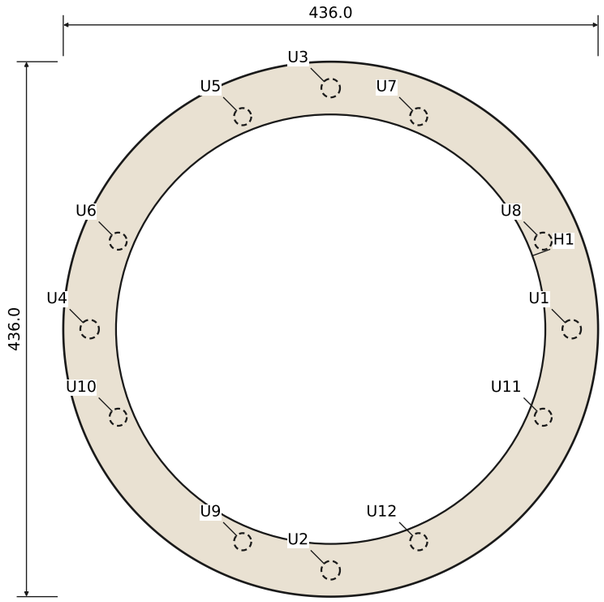
Sheet part: brace

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/woofer-ring.dxf. Marks listed at the right.



woofer-ring: 6 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/woofer-ring-underside.dxf (seen from that face)

mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	414.5	218.0	3.2 from inside
U2	Ø15.2	218.0	21.5	3.2 from inside
U3	Ø15.2	218.0	414.5	3.2 from inside
U4	Ø15.2	21.5	218.0	3.2 from inside
U5	Ø14.0	146.3	391.2	4 from inside
U6	Ø14.0	44.8	289.8	4 from inside
U7	Ø14.0	289.8	391.2	4 from inside
U8	Ø14.0	391.2	289.8	4 from inside
U9	Ø14.0	146.3	44.8	4 from inside
U10	Ø14.0	44.8	146.2	4 from inside
U11	Ø14.0	391.2	146.2	4 from inside
U12	Ø14.0	289.8	44.8	4 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

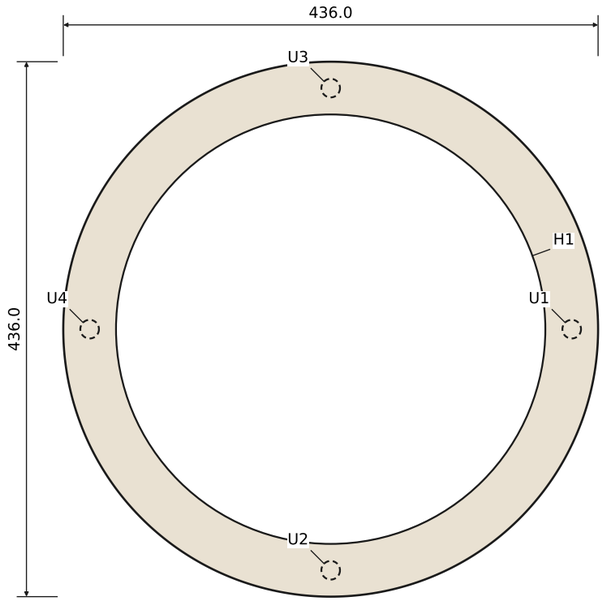
Sheet part: woofer-ring

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Scale 1:5. As the router sees it from the outer face; DXF: dxf/disc-frame.dxf. Marks listed at the right.



mark	size	x	y	depth
H1	Ø350.0	218.0	218.0	through
U1	Ø15.2	414.5	218.0	3.2 from inside
U2	Ø15.2	218.0	21.5	3.2 from inside
U3	Ø15.2	218.0	414.5	3.2 from inside
U4	Ø15.2	21.5	218.0	3.2 from inside

H: cut through. P: pocket from the face shown.
U: pocket from the inner face (dashed; its own DXF).
Centres from the lower-left corner,
seen from the outer face.

disc-frame: 12 mm mdf, 2 for the set, seen from its outer face

inner-face pockets: dxf/disc-frame-underside.dxf (seen from that face)

Sheet part: disc-frame

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

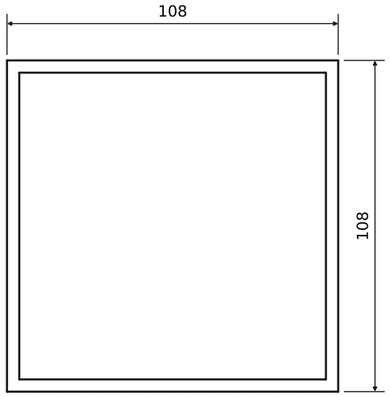
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

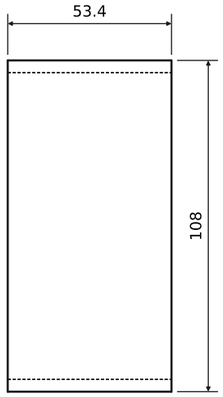
Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

8. port: asa print, on end; 89 cm3 each; 2 for the set; 1:2

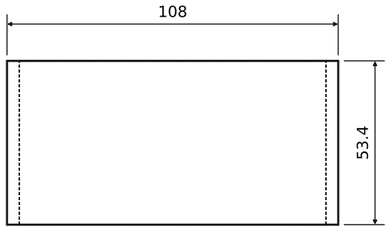
100 square, 53 long from the panel's face: tunes the box to 38 Hz; print 15 longer and trim to tune



FRONT



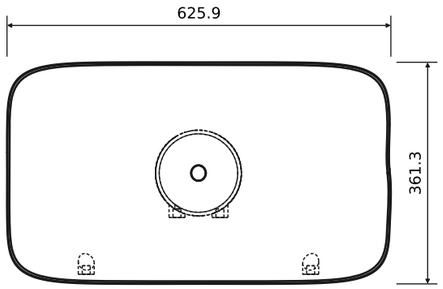
FROM THE RIGHT



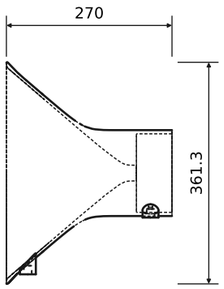
FROM ABOVE

11. waveguide: cast, cast-resin; 2293 cm3 each; 2 for the set; 1:10

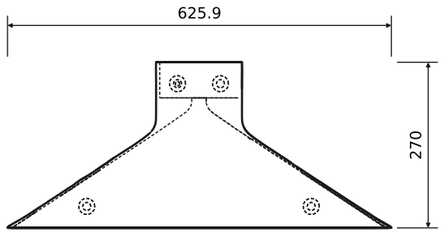
mouth 610 x 346, 200 deep, 1 in throat; a 10 rim rounded right round, 626 x 361 outside; a shell 6 thick following its flare; the driver in a pod 140 across behind, closed by a cap; four pads under it take the posts' brass inserts; cast in tinted translucent polyurethane in a two-part silicone mould from a CNC-cut plug, bead-blasted to frost



FRONT



FROM THE RIGHT



FROM ABOVE

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

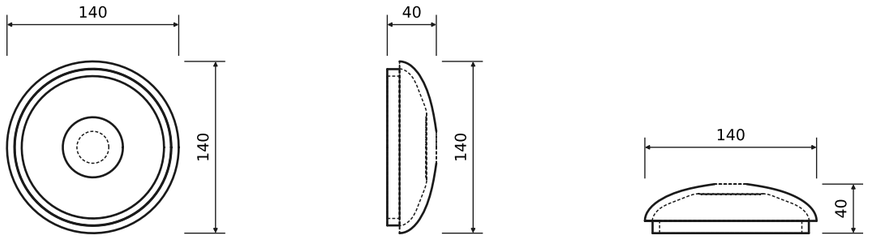
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

12. waveguide-cap: cast, cast-resin; 131 cm3 each; 2 for the set; 1:5

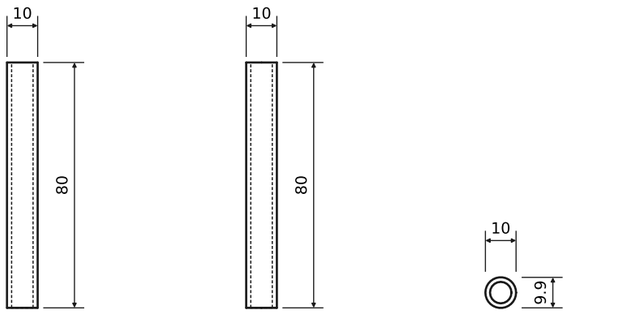
closes the pod behind the driver; a spigot locates it, three M4 screws hold it; the cable leaves through a post below



FRONT **FROM THE RIGHT** **FROM ABOVE**

14. post-1, post-2: lathe, stainless; 3 cm3 each; 4 for the set; 1:2

80 long (55 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top and 13 at the foot, each end into a brass insert



FRONT **FROM THE RIGHT** **FROM ABOVE**

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

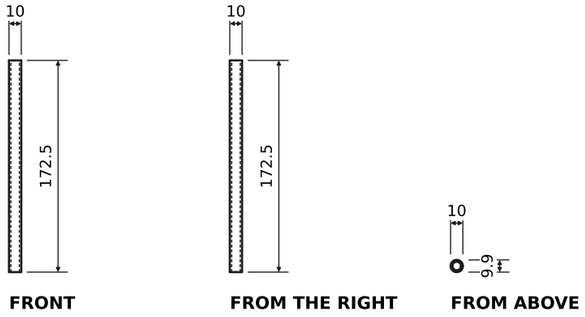
mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

Three views each, hidden edges dashed; STL and STEP files in stl/ and step/. Numbers are the parts list's.

15. post-3, post-4: lathe, stainless; 7 cm3 each; 4 for the set; 1:5

173 long (148 showing): stainless tube 10 x 7, threaded M10 x 1 12 at the top and 13 at the foot, each end into a brass insert; the HF cable runs down inside it



FRONT **FROM THE RIGHT** **FROM ABOVE**

Printed and machined parts

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15

For 2; prices are budgets (bom.md says where each comes from).

no.	item	make	qty	what	size	each USD
1	left, right	cut	4	18 mm Baltic birch plywood, BB/BB	420 x 880	26.80
2	top	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
3	bottom	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 420	15.96
4	front-panel	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
5	back	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 844	32.06
6	brace	cut	2	18 mm Baltic birch plywood, BB/BB	524 x 360	13.68
7	woofer-ring	cut	2	6 mm MDF	436 x 436	4.28
8	port	print	2	ASA, FDM (outdoor, paintable)	89 cm3	14.22
9	woofer	buy	2	FaitalPRO 15PR400-8 15 in woofer (treated paper cone, neodymium, 8 ohm)	393 frame, 356 cut-out, 165 deep	194.00
10	amp-plate	buy	2	Hypex FusionAmp FA122 DSP plate amplifier, 2 x 125 W into 4 ohm	315 x 120 x 55 plate, 291 x 96 recess	422.40
11	waveguide	cast	2	Tinted polyurethane casting resin, translucent	2293 cm3, 636 x 280 x 371	555.88
12	waveguide-cap	cast	2	Tinted polyurethane casting resin, translucent	131 cm3, 150 x 50 x 150	135.77
13	hf	buy	2	BMS 4550-16 compression driver, 1 in exit, polyester diaphragm, 16 ohm	123 across, 52 deep, 2 x M6 on 76.2	154.00
14	post-1, post-2	lathe	4	Stainless steel 304	20 x 20 x 90	16.08
15	post-3, post-4	lathe	4	Stainless steel 304	20 x 20 x 183	17.19
16	disc-frame	cut	2	12 mm MDF	436 x 436	4.28
17	cloth-disc	buy	2	Acoustically transparent speaker cloth (a knit sold for grilles), about 0.6 x 0.95 m a grille, in the finish's colour		[PRICE]
18	cloth-disc-magnets	buy	2	Grille magnets: eight 15 x 3 neodymium discs (N42), four attracting pairs, a set		[PRICE]
19	bark-face, bark	buy	6	Bark for laminating, a square metre: paper birch (Betula papyrifera) sheet from fallen or felled trees, or a smooth-barked gum's shed bark pressed flat (or a UV-printed veneer of it); approve a sample board first		[PRICE]
20	woofer-fixings	buy	2	M6 x 25 button heads (A2) with M6 T-nuts, a set of 8 (the 15PR400's eight holes on 375)		[PRICE]
21	hf-fixings	buy	2	M6 x 25 socket caps with washers (the compression driver to the horn's flange), a pair		[PRICE]
22	post-inserts	buy	2	Brass threaded inserts M10 x 1: four bonded into the waveguide's pads (13 across, 14 long) and four screwed into the cabinet's top (14 across, 15 long, one with a through-bore for the cable), a set of eight		[PRICE]

7 more lines: bom.md

Notes

1. Bark on the shell (sides, top, bottom and back): real birch bark sheet (Betula papyrifera, from fallen or felled trees) laminated to the plywood with PVA in a vacuum bag and sealed with a matt waterborne lacquer, wrapped round the shell's front edges; for eucalyptus, the shed bark of a smooth-barked gum (spotted or snow gum) pressed flat and laminated the same way, or a UV-printed veneer of it. Approve a sample board first.
2. Glue-up: sides to the bottom, the brace, then the front panel (set back 24 mm), the back, then the top.
3. The woofer flush in its rebate on a gasket, eight M6 T-nuts from inside, the ring over its frame; the port pushed in dry, trimmed to tune (impedance minimum at fb), then bonded.
4. The waveguide stands on four posts, stainless lamp tube 10 x 7 threaded M10 x 1 at both ends, screwed into brass inserts bonded in pads cast under it and into inserts in the top; the HF cable runs down inside a rear post.
5. The front: the panel faced in bark (1 mm) round the ring and the port, and a cloth disc over the woofer, its MDF frame standing on the ring on four magnet pairs, its face 18.5 clear of the cone at rest.

Parts list and notes

barkhorn: Two-way horn loudspeaker after Friendly Pressure, in birch bark or eucalyptus bark

mm | issue 2026-10-12 | first angle

Tolerances: sheet parts ±0.5, pockets ±0.2, holes +0.2/0, prints ±0.15